

A Chiral SU(2) Gauge-Sector Extension as a Candidate Effective Dark-Energy Component

Draft research note linking Yang-Mills local isotopic symmetry, Noether structure, and cosmological tests

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Source anchor: C. N. Yang and R. L. Mills, "Conservation of Isotopic Spin and Isotopic Gauge Invariance," Physical Review 96, 191-195 (1954).

Working status

This paper is a hypothesis-development document, not a claim that Standard Model weak SU(2) has already explained dark energy. The proposed path is to test whether a chiral SU(2)-like effective sector can mimic part of the late-time acceleration while satisfying SN, BAO, CMB-distance, isotropy, and parity-violation constraints.

Abstract

This note investigates whether a handedness-biased local SU(2) symmetry-breaking sector can be extended from particle-gauge mathematics into a cosmological effective component capable of contributing to late-time acceleration. Yang and Mills showed that making isotopic spin rotations local requires a compensating non-Abelian gauge field whose field strength contains a self-interaction term. That mathematical structure does not by itself imply dark energy. However, it supplies a rigorous framework in which a residual gauge condensate, chiral order parameter, or parity-biased vacuum sector can be parameterised and tested against cosmological data. The proposed first-stage implementation is phenomenological: add a chiral SU(2) effective density term to the existing Pantheon+/BAO/Planck likelihood program, then compare AIC/BIC and posterior constraints against Lambda-CDM, KdS, and FR variants. The second-stage implementation would derive the same term from an SU(2) field ansatz and compute its stress-energy tensor directly.

Motivation and core question

The core question is whether a handedness bias in local SU(2) symmetry breaking can leave a cosmological-scale remnant that behaves like some fraction of dark energy. The most conservative answer is: it can be modelled and tested, but it should not be asserted from Yang-Mills theory alone.

The original Yang-Mills paper establishes the general local-gauge mechanism: a global internal symmetry is promoted to a local symmetry, and a new gauge connection must be introduced to maintain covariance. The important feature for cosmology is that a non-Abelian gauge field has nonlinear self-interaction. In principle, such a sector can carry stress-energy and therefore enter the Friedmann equations.

The handedness component enters most naturally through modern chiral gauge theory, especially the weak-interaction structure $SU(2)_L \times U(1)_Y$. In that setting, left-handed fermions transform differently from right-handed fermions. The cosmological proposal is not that the ordinary weak interaction simply becomes dark energy, but that a residual or hidden SU(2)-like chiral gauge sector could act as an effective vacuum component.

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What Yang and Mills contribute mathematically

Yang and Mills consider isotopic spin rotations of a two-component field. If the isotopic orientation is permitted to vary independently at each spacetime point, the derivative of the field no longer transforms covariantly unless a gauge field is introduced. This is the local-gauge principle.

$$\psi(x) \rightarrow S(x) \psi(x)$$

Here $S(x)$ is a local 2×2 isotopic rotation. The gauge connection can be written schematically as $B_\mu = B_\mu^a T^a$, with T^a the SU(2) generators. The field strength is:

$$F_{mn} = \partial_m B_n - \partial_n B_m + g [B_m, B_n]$$

The commutator term is the defining non-Abelian feature. It means the gauge field interacts with itself and can carry the same kind of internal charge that it mediates. This is why non-Abelian gauge fields are plausible candidates for cosmological condensate models in a way that a simple scalar correction is not.

The Yang-Mills paper also shows that once the gauge field is included, the conservation structure is no longer a simple matter-current conservation law. The gauge field carries part of the conserved quantity. This is directly relevant to a Noether framing: local symmetry produces covariant conservation and constraint equations rather than a naive globally conserved current.

SU(2), handedness, and symmetry breaking

SU(2) is the group of 2×2 unitary matrices with determinant 1. Its Lie algebra has three generators, commonly represented by half the Pauli matrices. In the weak interaction, the crucial feature is that the gauge group acts chirally: SU(2)_L acts on left-handed fermion doublets.

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$$

This symmetry breaking, mediated by the Higgs mechanism in the Standard Model, gives mass to W and Z bosons and leaves the photon massless. It is strongly handed: left-handed and right-handed fermions do not have identical gauge roles.

However, the known electroweak scale is far too high to be identified directly with the present dark-energy scale. The useful research direction is therefore not ordinary electroweak vacuum energy as dark energy, but an effective late-time remnant, hidden sector, or emergent SU(2)-like field whose parameters can be fitted phenomenologically.

Ordinary SU(2) versus cosmological SU(2)-like sector

Issue	Standard Model SU(2) _L	Cosmological SU(2)-like proposal
Primary domain	Electroweak interactions of fermions and gauge bosons.	Effective late-time vacuum/gauge sector in FLRW cosmology.
Handedness	Built in: left-handed doublets transform under SU(2) _L .	Represented by chiral coupling, parity-odd term, or biased initial/order-parameter state.
Energy scale	Electroweak scale, much higher than dark-energy scale.	Must be tuned or dynamically suppressed to the observed late-time density.
Main observable	Particle interactions, W/Z masses, parity violation.	Expansion history, equation of state, anisotropy, CMB parity signatures.

Viability condition	Experimentally established particle theory.	Must compete with Lambda-CDM and satisfy cosmological constraints.
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Cosmological extension: field-theory form

A field-theory version can be expressed by coupling gravity to an SU(2) gauge field, a symmetry-breaking scalar/order parameter, and an optional parity-violating term. A schematic action is:

$$S = \int d^4x \sqrt{-g} \left[R/(16\pi G) - 1/4 F^a_{mn} F^{a mn} - 1/2 (D_m \Phi)^a (D^m \Phi)^a - V(\Phi) + \lambda_{\chi} \Phi F^a_{mn} \tilde{F}^{a mn} + L_m \right]$$

The term $F \tilde{F}$ is parity odd. It is the natural place to encode handedness bias. It does not automatically create dark energy, but it can bias left/right field configurations, source chiral gravitational-wave or CMB-polarization signatures, and alter the gauge-sector dynamics.

A single vector field usually breaks isotropy by selecting a direction. SU(2) has a useful escape route because the internal group index can be locked to the spatial index through an isotropic cosmic-triad ansatz:

$$A_i^a(t) = \phi(t) \delta_i^a, \quad A_0^a = 0$$

This configuration can preserve homogeneity and isotropy at the background level, making it a plausible testbed for an SU(2)-type cosmological energy component.

Phenomenological first-stage model

The fastest and safest implementation is not to solve the full SU(2) field equations immediately. Instead, introduce an effective density component $\Omega_{\chi}(z)$ into the existing Friedmann function and ask whether the data permit it.

$$E^2(z) = \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_k(1+z)^2 + \Omega_L + \Omega_{\chi}(z)$$

$$\Omega_{\chi}(z) = \Omega_{\chi 0} \exp \left[3 \int_0^z (1+w_{\chi}(z'))/(1+z') dz' \right]$$

Begin with the standard two-parameter equation-of-state form:

$$w_{\chi}(z) = w_0_{\chi} + w_a_{\chi} z/(1+z)$$

If w_0_{χ} is near -1 and w_a_{χ} is near 0, the component behaves like dark energy. If it strongly departs from this, it becomes a dynamical dark-energy or modified-sector model. This is the correct first diagnostic before claiming a deeper SU(2) origin.

Which program should be modified

The program to modify first is the existing cosmology likelihood/fitting program that already generates the plamb_runs folders and fits Pantheon+/SH0ES supernovae with BAO and optional Planck distance priors. Based on the current project structure, this is the same code path used for LCDM, KdS, FR, varM, varC, and acc-fluid runs.

Do not modify jwst_extract.py for this first test. That program is for retrieving JWST products and catalogues; it is not the correct location for a late-time expansion-history likelihood. JWST can become a later high-z diagnostic, but the first viability test belongs in the SN+BAO+Planck fitter.

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If there are several current scripts, select the one containing:

- the $E(z)$ or $H(z)$ model function;
- distance modulus $\mu(z)$ or luminosity-distance calculation;
- BAO $D_M(z)$, $D_H(z)$, $D_V(z)$, or $H(z)$ predictions;
- Planck distance-prior likelihood;
- AIC/BIC and run-folder saving into `plamb_runs`;
- command-line flags such as `--flat`, `--curved`, `--bao`, `--planck`, `--cc`, `--seed`, `--save-chain`, or similar.

That script should receive a new model branch, tentatively named `SU2_chiral_DE` or `FR_SU2_chiral`. The implementation should be kept modular so the phenomenological version can later be replaced by a derived SU(2) field-equation version.

Proposed code modifications

Code area	Modification	New parameters	Reason
Model registry / CLI	Add <code>--model SU2_chiral_DE</code> and optional <code>--su2-ansatz phenomenological triad</code> .	<code>model</code> , <code>su2_ansatz</code>	Keeps the extension comparable with Λ CDM, KdS , and FR.
Parameter vector	Add bounded parameters for the effective chiral sector.	<code>Omega_chi0</code> , <code>w0_chi</code> , <code>wa_chi</code> , <code>chi_bias</code>	Allows likelihood exploration without hard-coding a result.
$E(z)$ function	Add $\Omega_{\chi}(z)$ to the Friedmann function.	$\Omega_{\chi}(z)$	Directly tests expansion-history effect.
Priors	Add conservative priors and optional flatness closure.	bounds on <code>Omega_chi0</code> , <code>w0_chi</code> , <code>wa_chi</code>	Prevents the new sector from absorbing unphysical parameter space.
Output tables	Save fitted SU2 parameters, derived $w(z)$, AIC, BIC, <code>chi2</code> blocks.	<code>w0</code> , <code>wa</code> , <code>Omega_chi0</code> , $\Delta\text{AIC/BIC}$	Makes comparison against existing <code>plamb_runs</code> automatic.
Diagnostics	Plot $w_{\chi}(z)$, $\Omega_{\chi}(z)$, residuals, and posterior corner.	derived arrays	Separates dark-energy-like behaviour from parameter degeneracy.
Future field version	Add triad stress-energy calculation behind the same interface.	<code>g_su2</code> , <code>phi0</code> , <code>phidot0</code> , <code>V0</code> , <code>lambda_chi</code>	Moves from phenomenology to a physical SU(2) model.

Recommended first parameter set

For the first run, avoid an over-parameterised theory model. Use a minimal effective component:

- `Omega_chi0`: present-day fractional density of the chiral SU(2)-like component.
- `w0_chi`: present-day equation of state.
- `wa_chi`: redshift evolution of equation of state.
- `chi_bias`: optional bookkeeping parameter for handedness/parity bias, initially not coupled to SN/BAO distances unless a parity dataset is added.

Suggested initial bounds:

Parameter	Initial bound	Comment
<code>Omega_chi0</code>	0 to 1	Use with closure condition if replacing part of Ω_{Λ} .
<code>w0_chi</code>	-1.5 to -0.3	Allows phantom-like and quintessence-like behaviour without excessive

		freedom.
wa_chi	-2 to +2	Broad enough for first pass; tighten after stability checks.
chi_bias	-1 to +1	Diagnostic parity-bias amplitude; not distance-active initially.

10. Implementation algorithm

1. Clone the current SN+BAO+Planck fitter into a new script or branch before editing.
2. Add a model name SU2_chiral_DE to the model-selection switch.
3. Extend the parameter vector builder so Omega_chi0, w0_chi, and wa_chi can be active parameters.
4. Implement w_chi(z) and Omega_chi(z) as separate pure functions.
5. Modify E2_model(z, params) so the chiral term is added only for the SU2 branch.
6. Run LCDM first to confirm no baseline drift.
7. Run SU2_chiral_DE on SN-only, SN+BAO, and SN+BAO+Planck.
8. Save run metadata, parameter bounds, best-fit values, covariance/chain files, residual plots, and AIC/BIC.
9. Compare against existing LCDM, KdS, FR_varM, FR_varC, and acc-fluid runs.
10. Only after a statistically meaningful result should the field-theory triad version be implemented.

11. Pseudocode insertion

```
def w_chi_z(z, p):
    a = 1.0 / (1.0 + z)
    return p.w0_chi + p.wa_chi * (1.0 - a)

def omega_chi_z(z, p):
    # CPL closed form for w(z) = w0 + wa*z/(1+z)
    a = 1.0 / (1.0 + z)
    exponent = -3.0 * p.wa_chi * (1.0 - a)
    return p.Omega_chi0 * a ** (-3.0 * (1.0 + p.w0_chi + p.wa_chi)) * exp(exponent)

def E2_model(z, p):
    E2 = p.Omega_m * (1+z)**3 + p.Omega_r * (1+z)**4 + p.Omega_k * (1+z)**2
    if p.model == "SU2_chiral_DE":
        E2 += omega_chi_z(z, p)
        if p.include_lambda:
            E2 += p.Omega_L
    else:
        E2 += existing_dark_sector_terms(z, p)
    return E2
```

12. Acceptance tests

The model should not be promoted beyond exploratory status unless it passes the following checks:

- Baseline reproducibility: LCDM and existing FR/KdS results are unchanged after the code refactor.
- Numerical stability: E2(z) remains positive over the full integration grid.
- Distance consistency: SN, BAO, and Planck likelihood blocks remain separately reported.
- Information criteria: DeltaAIC/DeltaBIC are reported against the exact same data combination.
- Parameter identifiability: posterior constraints are not purely prior-bound.
- Physical plausibility: the inferred w_chi(z) is close enough to -1 at low redshift to qualify as dark-energy-like.

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- Isotropy: any later vector/gauge-field version must avoid large anisotropic stress.
- Parity tests: handedness should eventually be checked with CMB TB/EB, cosmic birefringence, or chiral gravitational-wave observables rather than inferred from expansion data alone.

13. Noether interpretation

The Noether framing is useful but must be stated carefully. A global continuous symmetry gives an ordinary conserved current. A local gauge symmetry gives gauge redundancy, constraints, and covariant conservation. In Yang-Mills theory the matter current alone is not the full conserved structure; the gauge field itself carries part of the charge/current.

$$D_\mu J^\mu = 0$$

For the cosmology project this suggests a clean conceptual framing: local SU(2)-type symmetry breaking may redistribute conserved structure between matter fields, gauge fields, and vacuum/order-parameter degrees of freedom. If the residual sector has a late-time equation of state near -1, it can be tested as an effective dark-energy component.

14. Interpreting possible outcomes

Outcome	Meaning	Next action
No AIC/BIC improvement; parameters prior-bound	The SU2 effective sector is not supported by distance data.	Stop or keep only as a theoretical appendix.
Small AIC improvement but BIC penalty	May be absorbing mild residuals but not worth extra complexity.	Check residuals and degeneracies; do not overclaim.
Clear improvement with w near -1	Potential dark-energy-like phenomenology.	Run robustness checks, full chains, and dataset splits.
Improvement only when Planck is removed	Could be easing H0/SN tension rather than explaining dark energy.	Compare SN-only, BAO-only, and Planck-prior pulls.
Parity signal required but absent	Handedness component unsupported observationally.	Separate expansion-history sector from parity-bias claim.

15. Recommended run matrix

Run	Data	Geometry	Model	Purpose
R1	SN only	Flat	ΛCDM vs SU2_chiral_DE	Check whether SN residuals favour the new component.
R2	SN + BAO	Flat	ΛCDM/KdS/FR/SU2	Main first comparison against current project models.
R3	SN + BAO + Planck	Flat	ΛCDM/KdS/FR/SU2	High-constraint viability test.
R4	SN + BAO	Curved	SU2_chiral_DE	Check degeneracy with curvature.
R5	SN + BAO + Planck	Curved	SU2_chiral_DE	Final distance-prior robustness test.
R6	Parity data later	Flat	SU2 + chi_bias	Only after expansion-history viability is established.

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16. Conclusion

The Yang-Mills SU(2) local-gauge mechanism provides a mathematically serious foundation for considering non-Abelian gauge sectors in cosmology. A handedness-biased SU(2)-like symmetry-breaking remnant could, in principle, be parameterised as an effective dark-energy component. The viable research claim is not that ordinary Standard Model SU(2) explains dark energy, but that an SU(2)-inspired chiral gauge sector can be tested as a controlled extension of the existing FR/KdS cosmology framework.

The immediate technical step is to modify the existing Pantheon+/BAO/Planck fitting program, adding a model branch for SU2_chiral_DE with $\Omega_{\chi 0}$, $w_{\chi 0}$, w_a , and a dormant χ_{bias} parameter. The model should then be ranked against LCDM, KdS, and FR using the same data cuts, priors, integration settings, AIC/BIC reporting, and run-folder audit trail already used in the project.

References

Yang, C. N., and Mills, R. L. (1954). Conservation of Isotopic Spin and Isotopic Gauge Invariance. *Physical Review*, 96, 191-195. Uploaded source PDF reviewed in this session.

Standard comparison targets in the current project: LCDM, KdS, FR variable-c, FR variable-M, and acc-fluid model branches using Pantheon+/SH0ES, BAO, and Planck distance-prior likelihoods.

17. July 14, 2026 empirical diagnostic addendum

Purpose. This addendum updates the SU(2) gauge-sector proposal after the July 14 diagnostic runs across supernova raw-MU calibration, Quaia anisotropy, HHT residual screening, and SPARC rotation-curve posterior checks. The main change is evidential rather than theoretical: the model remains a testable effective-component hypothesis, but the current diagnostics do not support discovery or full-sample promotion language.

17.1 Claim boundary after July 14 diagnostics

- Do not state that a stable SU(2)-associated angular mode has been detected in Quaia.
- Do not promote the SPARC filtered-family result as a full-sample PLAMB win.
- Use the hierarchical raw-MU offset likelihood as the default supernova calibration diagnostic.
- Use the pre-registered zHD external-flow frame as the default redshift-frame readout; keep zHEL and zCMB as controls.
- Treat HHT/CMB/SN structure as candidate-feature and method-development evidence until new surrogate and window controls pass.

17.2 Empirical status table

Diagnostic	July 14 result	Manuscript interpretation
Hierarchical raw-MU offset likelihood	Pantheon+SH0ES, DES-Dovekie raw, and Union3.1 were fit with dataset and survey calibration offsets. The best HD row used moderate dataset+survey priors with BIC-like score 3147.056. The beta-free H0-fixed row gave beta = 0.521314.	This is now the default SN diagnostic. The improvement is calibration-mediated, not standalone evidence for a new cosmology.
External-flow redshift audit	Primary zHD fits favored LambdaCDM over the exact-log kernel: Pantheon+ deltaAIC(log-LCDM) = +3.202 and DES-Dovekie = +1.483. The exact-log preference fades along the zHEL-to-zHD alpha path.	Use zHD as the headline frame. zHEL and zCMB are controls unless a physical reason for another frame is specified before fitting.
SU2/Quaia global look-elsewhere scan	Strongest row: z=1.00-1.50, b > 35, formal SNR = 3.70594. Global p-values were ordinary: p(max SNR) = 0.418605 and p(max SU2-priority) = 0.166113.	Fails as a full-scan global anisotropy detection. Keep only as a targeted follow-up hook.
SU2/Quaia direction and targeted stability	The stable lower-side window is z=0.95-1.45, with SNR = 4.785 and direction spread 13.47 deg in targeted mocks. Joint p = 0.000666. The narrower z=1.15-1.35 window gives joint p = 0.005330.	Interesting targeted evidence, but not a discovery claim because the global look-elsewhere gate did not pass.
Quaia template regression and full mock catalogs	Smooth Galactic quadrupole terms did not erase the dipole, but photometry/depth/proper-motion controls rotated the direction and suppressed high-latitude amplitudes. Window-matched object mocks gave z=1.00-1.50 p_amp about 0.042-0.044, while CMB-projected p-values were about 0.189-0.191.	External dust, depth, scanning-law, and ecliptic/Galactic templates are required before physical interpretation.
SPARC filtered-family posterior	RAR convergence was rescued: max Rhat improved from 1.300 to 1.031. PLAMB aggregate all-Q2 delta became +28.941 and low-acceleration outer points +48.508, but high inclination remained -358.865 and gas dominated remained -28.175.	SPARC is subset wins, not a full-sample PLAMB win. High-inclination and gas-dominated failures are explicit gating failures.
HHT/CMB/SN residual screening	SN residual weighted RMS was LCDM 0.15159, SU2 0.15548, SU2R 0.15542. Extended SU2-minus-LCDM residual IMF energy was not unusual under shuffled controls: max-energy p = 0.9703 and sum-energy p = 0.9604.	Archive as method development and candidate-feature evidence, not as confirmed CMB-SN HHT coupling.

17.3 Updated acceptance gates

Gate	Required before stronger claim language
Supernova likelihood	Replace fixed prior grids with a transparent sensitivity analysis tied to external calibration budgets. Keep zHD as the default release-frame readout.
Quaia angular mode	A stable $z \sim 1.0$ -1.5 or pre-registered subwindow mode must survive global look-elsewhere correction, latitude-cut perturbations, and external dust/depth/scanning-law templates.
Multipole coherence	Show a coherent dipole/quadrupole or multipole pattern rather than one marginal dipole-like row.
SU2 scalar link	Demonstrate that angular-mode strength relates to the SU2 scalar residual curve better than survey-template explanations.
SPARC rotation curves	Reverse high-inclination and gas-dominated posterior-predictive failures, or preregister a physical exclusion rule before viewing those outcomes.
HHT residual structure	Run probe-separated, pre-registered surrogate controls and survey-window controls before interpreting IMF structure physically.

17.4 Updated conclusion paragraph

The July 14 diagnostics sharpen the research program rather than closing it. The SU(2) gauge-sector proposal remains mathematically motivated and now has several concrete empirical hooks, especially the targeted Quaia $z=0.95$ -1.45 window and the calibration-aware raw-MU likelihood. However, the global look-elsewhere audit, template sensitivity, redshift-frame dependence, HHT surrogate controls, and SPARC stress failures all argue for disciplined exploratory language. The appropriate current claim is that the SU(2)/PLAMB framework defines a testable suite of diagnostics with targeted anomalies and clear falsification gates, not that it has displaced LambdaCDM or established a physical SU(2) angular symmetry.

17.5 Archived result sources

- github_export/results/2026-07-14/rawmu/joint_rawmu_hierarchical_and_redshift_preregistered_audit_2026-07-14.md
- github_export/results/2026-07-14/su2_quaia/su2_quaia_global_lookelsewhere_readout_2026-07-14.md
- github_export/results/2026-07-14/su2_quaia/su2_quaia_targeted_stability_readout_2026-07-14.md
- github_export/results/2026-07-14/su2_quaia/su2_quaia_template_regression_readout_2026-07-14.md
- github_export/results/2026-07-14/sparc/sparc_filtered_rar_rescue_readout_2026-07-14.md
- github_export/results/2026-07-14/hht_cmb_sn/hht_cmb_sn_scan_readout_2026-07-14.md